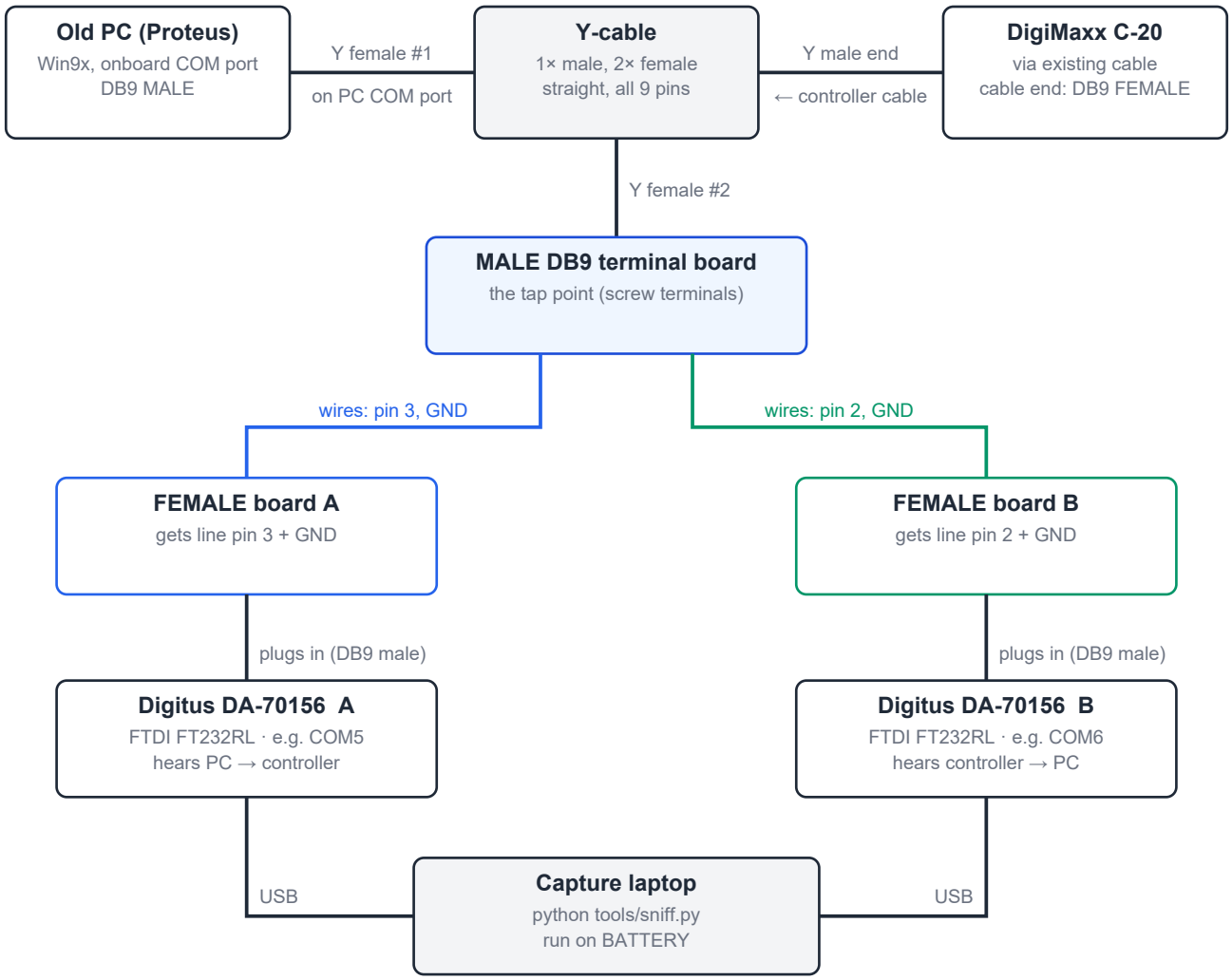
RS-232 Passive Tap — Wiring Guide

Listening to both directions of the Proteus PC ↔ FORM+TEST DigiMaxx C-20 serial link, without interrupting it. University of Patras · 4000 kN compression press.

How it works

RS-232 uses one wire per direction: **pin 3** carries the PC's commands and **pin 2** carries the controller's replies (numbering at the PC connector). The Y-cable exposes every line on a spare connector. A male terminal board breaks those lines out to screw terminals, and only the two data lines plus ground are wired onward, each to the **receive input** of its own USB adapter. The adapters' outputs are never connected, so the tap is electrically listen-only.

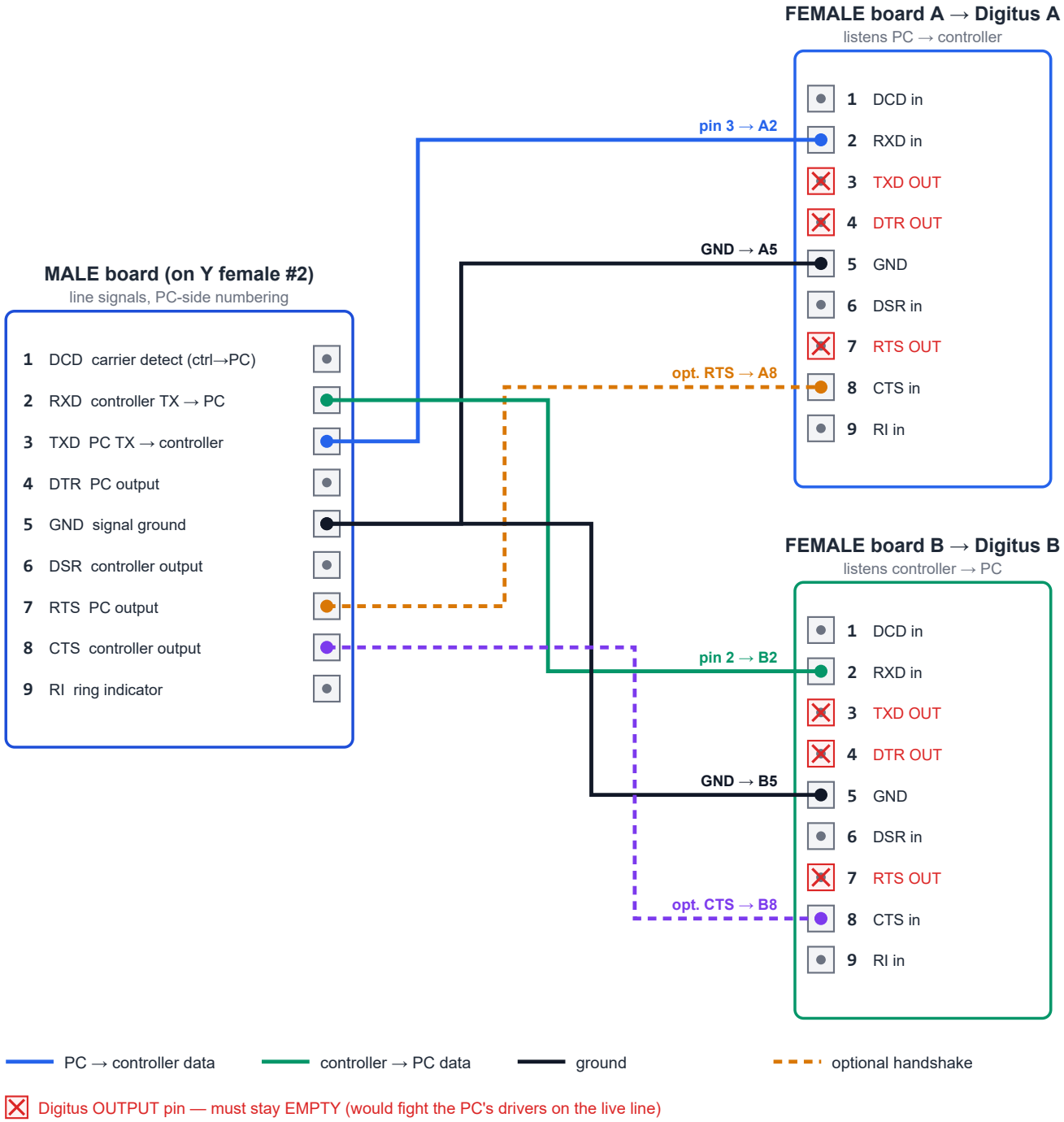


Orientation matters. The Y-cable must sit at the **PC end** of the link (Y female #1 directly on the PC's COM port). All pin numbers in this guide are as seen at the PC connector, regardless of how the controller's own cable is wired internally.

Terminal wiring

Four wires are required; two more are optional and only add handshake-line logging (`--lines`). Two wires share the male board's GND terminal (or use its extra GND terminal, if present).

> **Never connect pins 3, 4 or 7 on boards A or B.** Those are the Digitus adapters' TXD, DTR and RTS *outputs*. Connected to the tap they would drive the same wires as the PC and corrupt the command stream while the press is under load. Software cannot switch them off.



Wire list

#	From: MALE board	To	Carries	Required
1	pin 3 (TXD)	board A pin 2 (RXD in)	PC → controller data	yes
2	pin 2 (RXD)	board B pin 2 (RXD in)	controller → PC data	yes
3	pin 5 (GND)	board A pin 5 (GND)	signal ground	yes
4	pin 5 (GND)	board B pin 5 (GND)	signal ground	yes
5	pin 7 (RTS)	board A pin 8 (CTS in)	PC RTS handshake	optional
6	pin 8 (CTS)	board B pin 8 (CTS in)	controller CTS handshake	optional

Parts list

#	Part	Qty	Status	Notes
1	**MEIRIYFA DB9 Y-splitter**, 1× male → 2× female, straight-through, 50 cm	1	ordered	Amazon B0DQW2H1CC. Verify 1:1 continuity on all 9 pins before use.
2	**Digitus DA-70156** USB 2.0 → RS-232, FTDI FT232RL, DB9 male	2	have	Set Latency Timer to 1 ms (Device Manager → Port Settings → Advanced).
3	**DB9 MALE → screw-terminal board** (solderless breakout, 3.96 mm terminals)	1	to buy	Plugs into Y female #2. Local: GRobotronics (grobotronics.com).
4	**DB9 FEMALE → screw-terminal board**	2	to buy	Digitus A and B plug into these. Prefer versions with a case.
5	Hook-up wire, 22 AWG, stranded, ~15 cm pieces	4 (+2)	to buy	Use different colours for pin 3, pin 2 and GND.
6	Capture laptop, Python 3 + pyserial, 2 free USB ports	1	have	Run on battery during capture.
7	Multimeter with continuity beeper	1	—	For the pre-connection checks.
8	Slotted screwdriver, 2 mm	1	—	For the terminal screws.

Easiest purchase for items 3–4: two **male + female** breakout packs, e.g. YIOVVOM DB9 breakout M+F with case (Amazon B07DKZCL5B) or Twinklebay DB9 breakout M+F (Amazon B073RGGDKH). That yields 2 male + 2 female boards; the extra male board is a spare.

DB9 pinout reference

Pin	Signal	On the tapped line (PC connector)	On the Digitus (DTE)	Tap it?
1	DCD	carrier detect (ctrl→PC)	DCD in	no
2	RXD	controller TX → PC	RXD in	yes → B2
3	TXD	PC TX → controller	TXD OUT	yes → A2
4	DTR	PC output	DTR OUT	no
5	GND	signal ground	GND	yes → A5, B5
6	DSR	controller output	DSR in	no
7	RTS	PC output	RTS OUT	optional → A8
8	CTS	controller output	CTS in	optional → B8
9	RI	ring indicator	RI in	no

Checks before touching the press

1	**Y-cable continuity.** With nothing plugged in, confirm each of pins 1–9 on the male end beeps through to the same pin on *both* female ends, and no pin shorts to a neighbour.
2	**Tap assembly continuity.** With the Digitus adapters unplugged, confirm only: male 3 ↔ A2, male 2 ↔ B2, male 5 ↔ A5 and B5 (plus the optional 7 ↔ A8, 8 ↔ B8). A3, A4, A7, B3, B4 and B7 must connect to **nothing**.
3	**Baseline.** Insert the Y-cable between PC and controller cable, with nothing on Y female #2. Start Proteus and confirm the status bar shows **HARDWARE OK**.
4	**Add the tap.** Plug the male board into Y female #2. Confirm HARDWARE OK still shows.
5	**Add the adapters.** Plug Digitus A and B into boards A and B, USB into the laptop (on battery). Note their COM numbers and set Latency Timer = 1 ms on both. Confirm HARDWARE OK once more.
6	**Rollback** at any sign of trouble: unplug the Y-cable and reconnect the controller cable directly to the PC.

Capture

Find the baud rate first (keep Proteus' measurement window open so it is polling), then record sessions. Type a short note + Enter during capture to mark what you did (e.g. *clicked Zero*).

```
python tools/sniff.py --list

python tools/sniff.py --scan --pc-port COM5 --ctrl-port COM6

python tools/sniff.py --pc-port COM5 --ctrl-port COM6 --baud 9600

python tools/sniff.py --pc-port COM5 --ctrl-port COM6 --baud 9600 --lines   # with optional wires
```

If replies show up labelled `PC>CTRL`, the two COM numbers are swapped.

Safety notes

The tap is listen-only by design. Do not wire any Digitus output and do not send anything from the laptop to the controller at this stage.
Run the laptop on battery. Its ground is joined to the press's signal ground through the tap; a mains-powered laptop can create a ground loop and inject noise.
Do not use Proteus' **Calibrate** menu while capturing until the protocol is understood — it may write to the controller.
The controller's hardware STOP and the press emergency stop remain the primary safety devices at all times.